

The Geometry of Admissibility

Quotients, Lattices,
Sheaves, and Functorial
Reachability in AI Planning

A visual synthesis of the mathematical framework
governing state space representations.

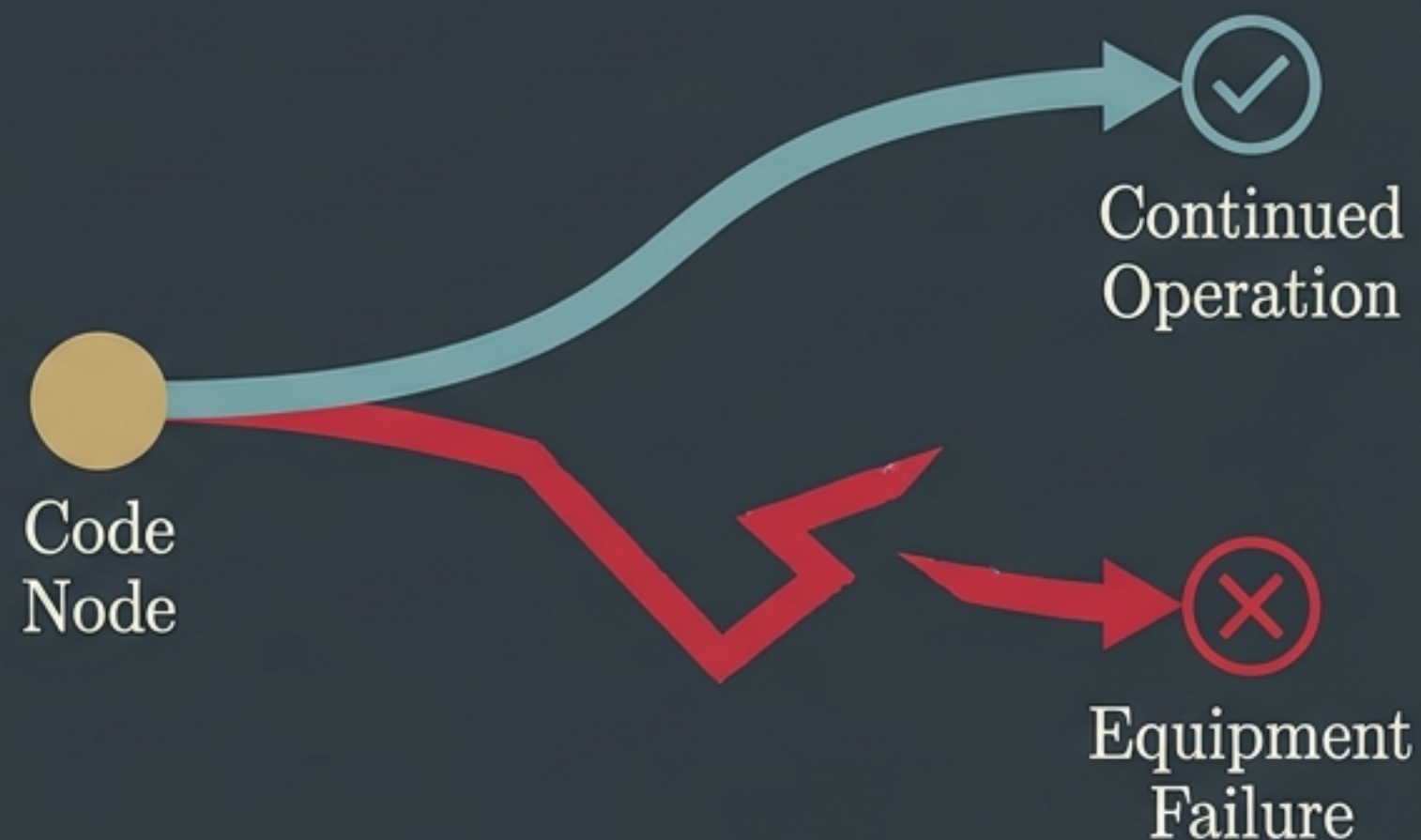


The Pull of Compression



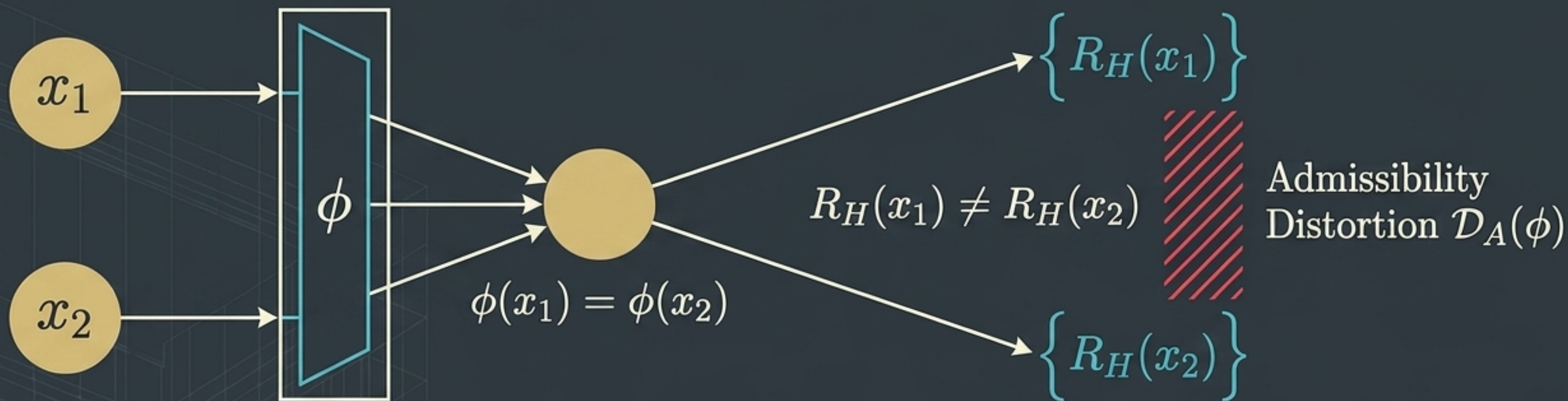
Standard encoders collapse visually similar states. A cracked insulator and a safe one become statistically indistinguishable.

The Requirement of Planning



Planning breaks when identical codes have radically different operational futures. This is the root of planning failure.

Defining Admissibility & Distortion



Admissibility Equivalence ($\sim_A$)

States x_1 and x_2 are equivalent only if their H-step reachability sets are identical.

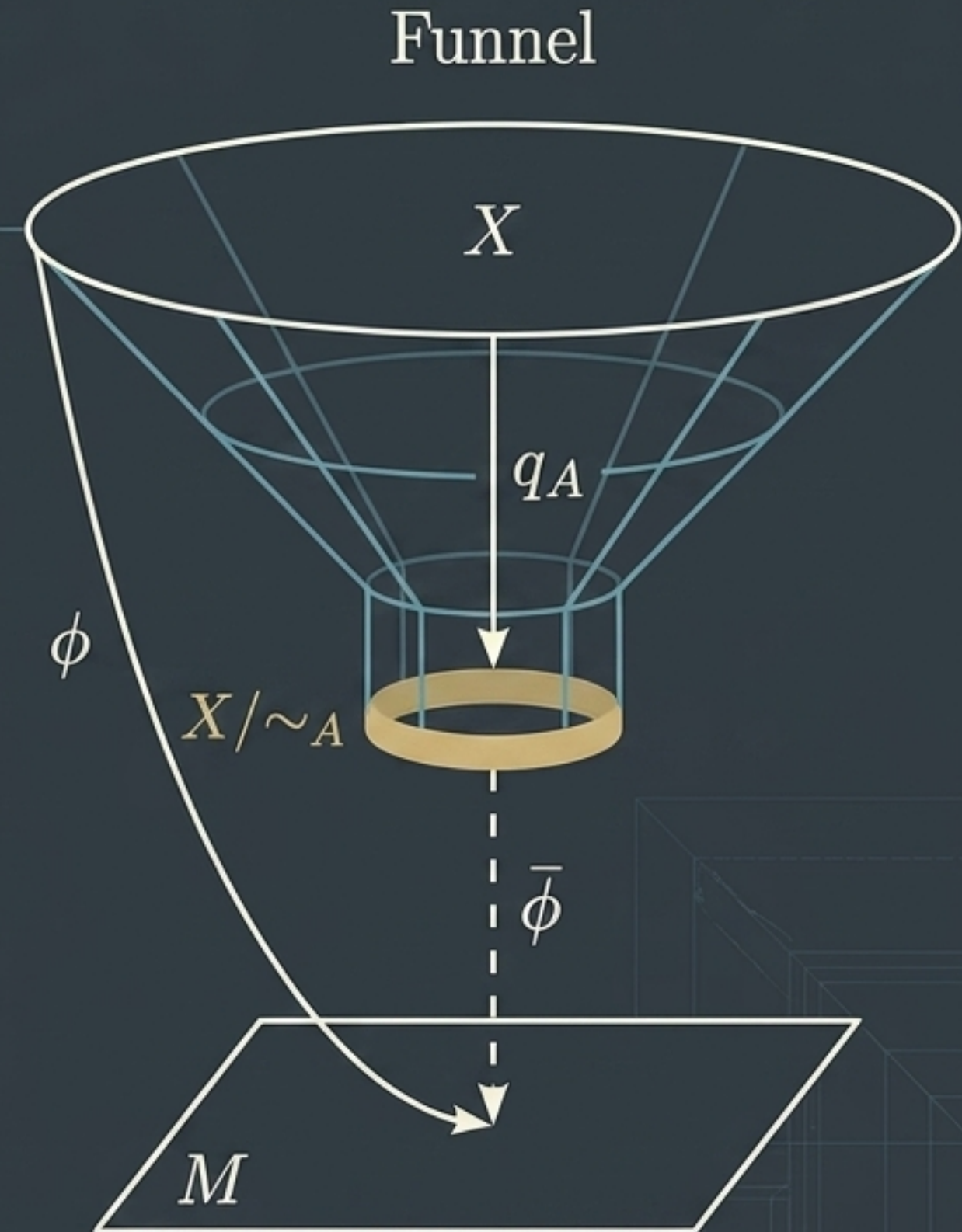
Admissibility Distortion

The penalty incurred—measured as reachability separation—when an encoder collapses states that are not admissibility-equivalent.

Pillar 1: The Category of Admissibility (**Adm**)

Key Insights

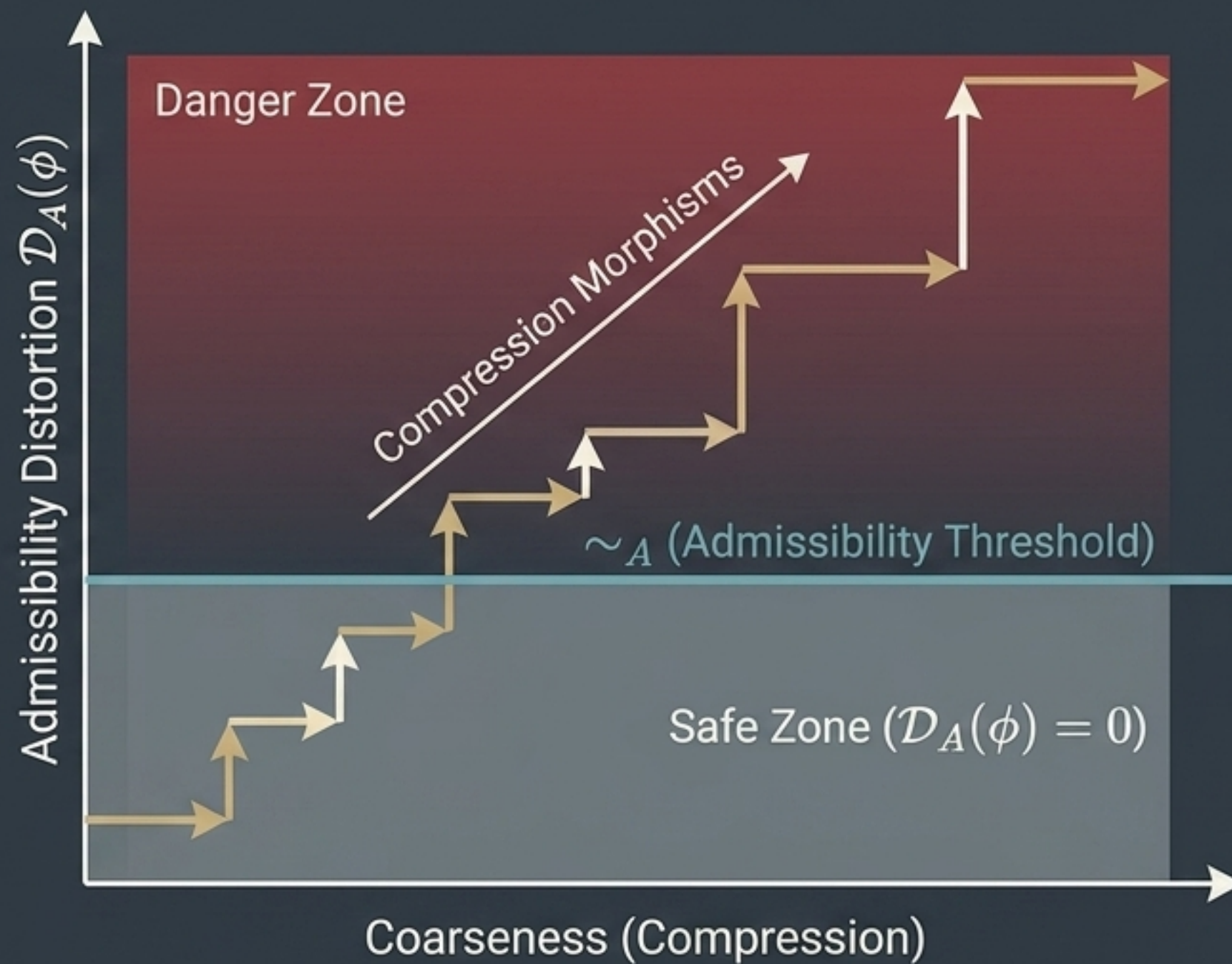
- q_A is the initial object: the “most compressed” admissible representation.
- Every class-constant map ϕ must factor through q_A .
- An encoder is only admissible if its unique factor $\bar{\phi}$ is strictly injective.



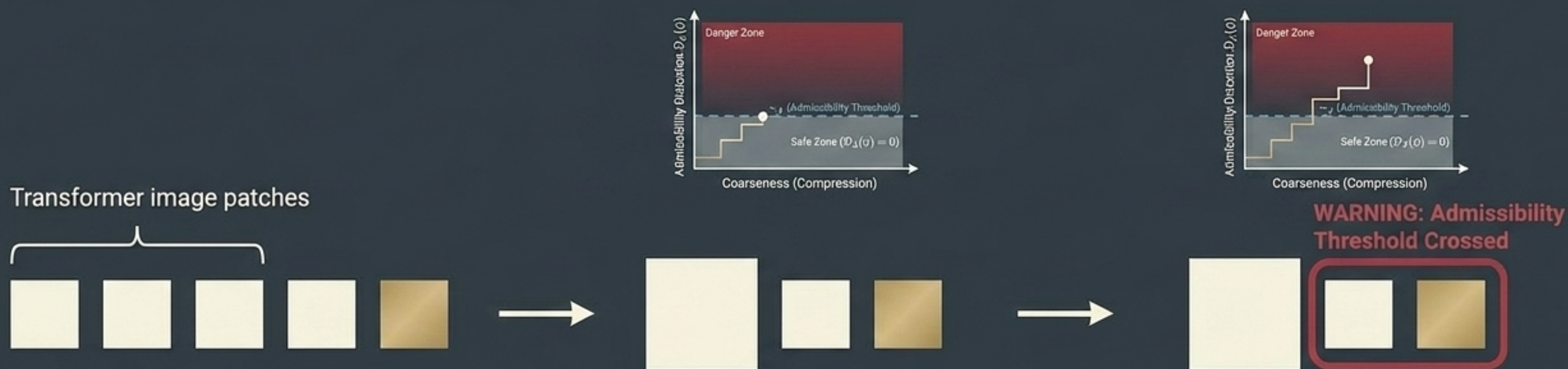
Pillar 2: The Quotient Lattice $\mathcal{L}(X)$

Distortion as a Height Function

Distortion is a monotone height function. Every compression step upward through the lattice toward coarser equivalence relations guarantees an increase in admissibility distortion once the $\sim_A$ threshold is crossed.



Lattice Movement in Practice: Token Pruning



The Action

Discarding or merging visual tokens induces an equivalence relation ($\sim_{\text{prune}}$).

The Consequence

Pruning is literally upward movement in the quotient lattice $\mathcal{L}(X)$.

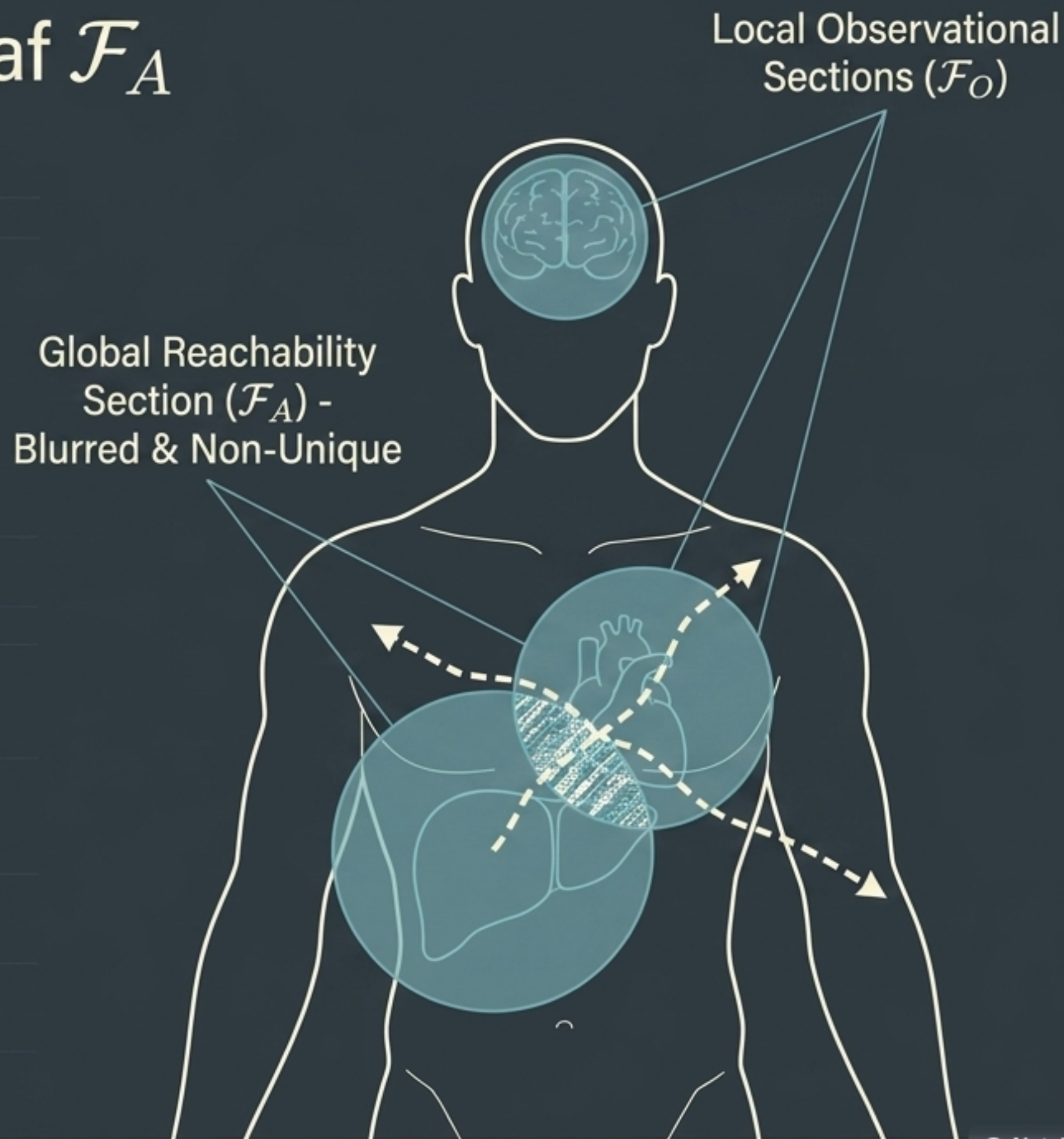
The Goal

Compute importance scores to approximate admissibility-aware quotienting—collapsing only tokens whose reachable futures are identical.

Pillar 3: The Reachability Sheaf $\mathcal{F}_A$

The Gluing Problem

- **Locality:** Passive observation only provides local sections (e.g., individual organ biological age).
- **Failure to Stitch:** Compatible local observations fail to stitch together into a single, unique global trajectory (e.g., whole-body biological age). The overlap is mathematically undefined.



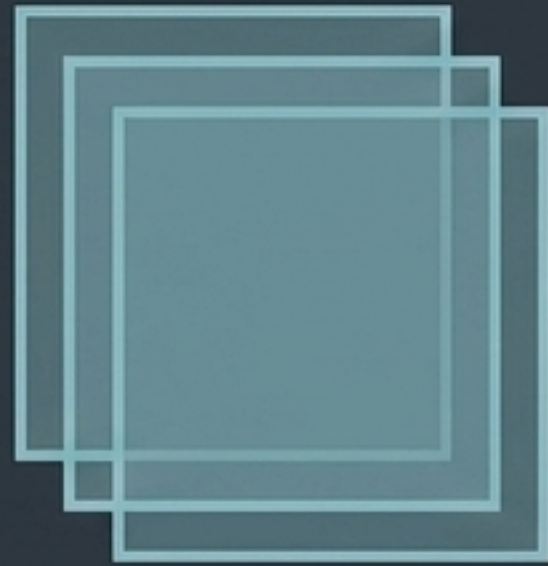
The Observational vs. Interventional Divide

	Observation Presheaf ($\mathcal{F}_O$)	Reachability Sheaf ($\mathcal{F}_A$)
Topology	Coarser (Fewer open sets)	Finer (Alexandrov topology τ_A)
Sections	Local (Spotlight patches)	Global (Complete system map)
Data Source	Passive observation sequences	Active, policy-driven reachability
Equivalent Example	States s_0^0 and s_0^1 have identical passive distributions	$s_0^0 \rightarrow s_{\text{safe}}$ while <u>$s_0^1 \rightarrow s_{\text{danger}}$</u>

Takeaway: Observational equivalence and admissibility equivalence are fundamentally distinct mathematical relations.

The Sheaf Gluing Failure

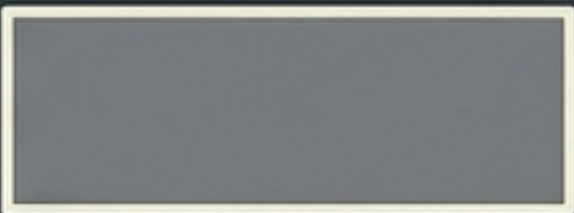
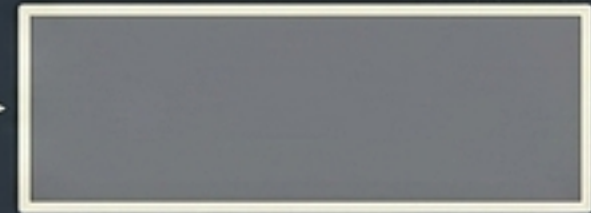
Input: Compatible Local
Sections $s_i \in \mathcal{F}_0(U_i)$



Gluing Mechanism



Global Section 1

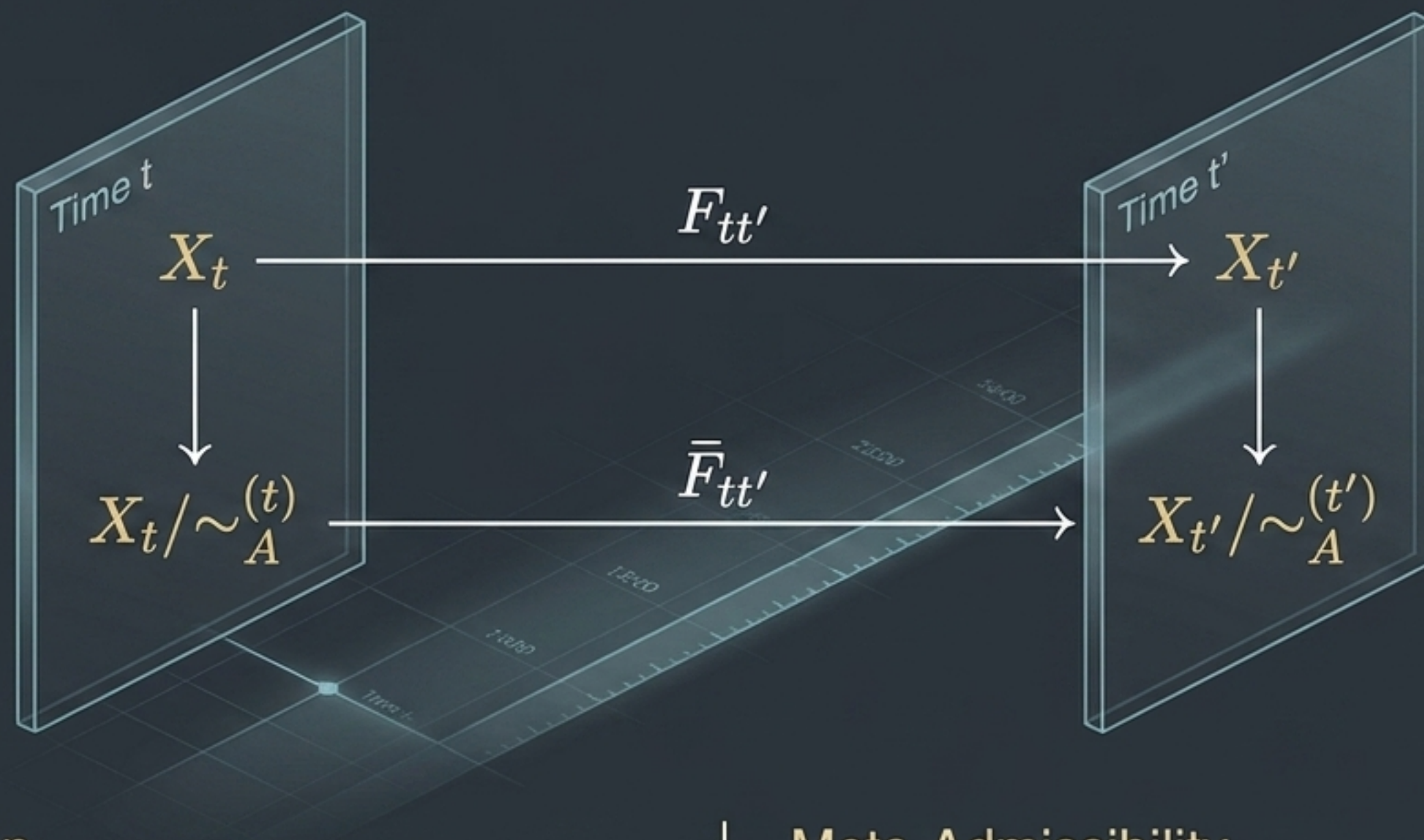


Global Section 2

Theorem Summary

The global reachability section is not uniquely determined by any complete system of local observational sections. Passive data is equally consistent with fundamentally distinct operational futures.

Pillar 4: Temporal Functors & Meta-Admissibility



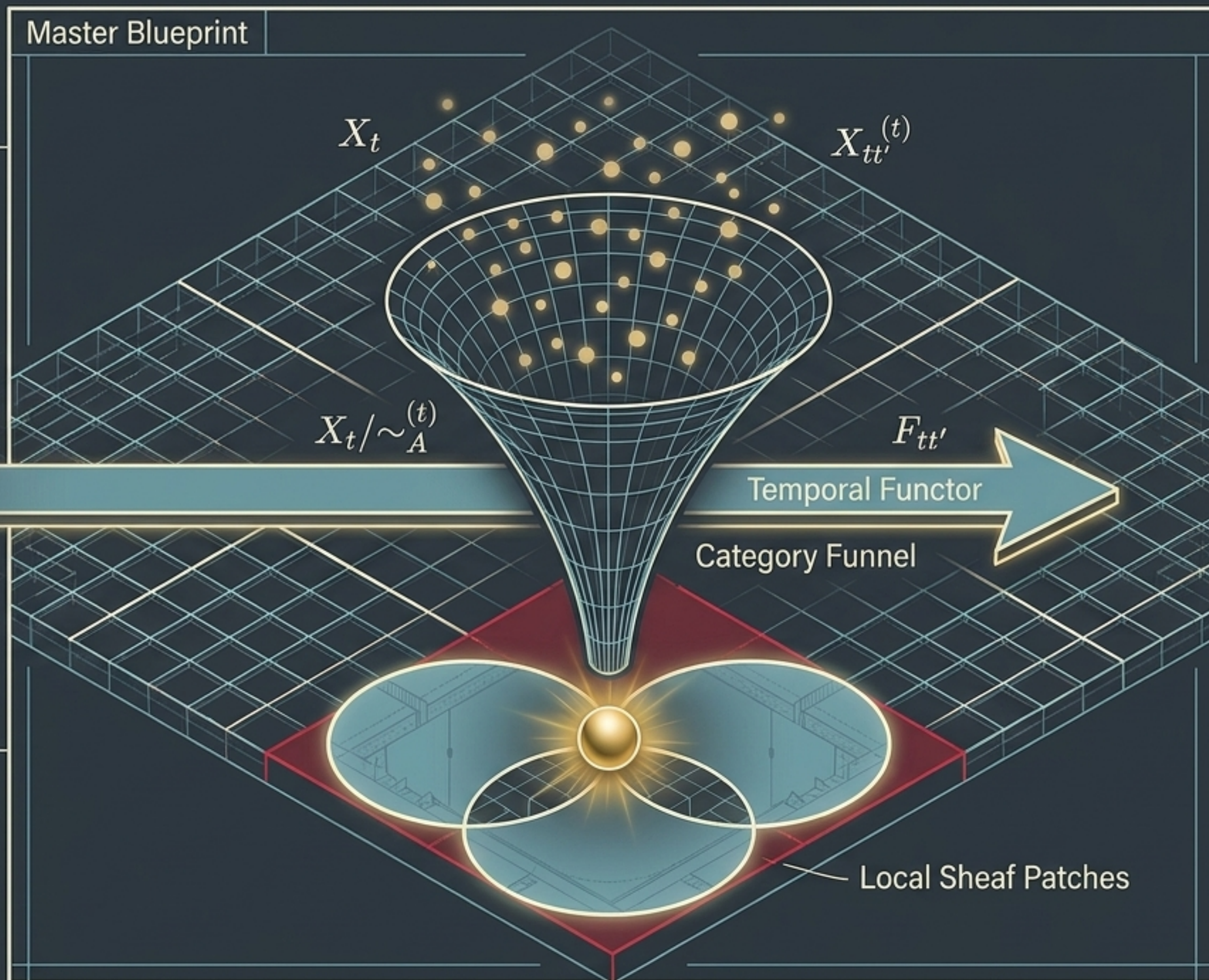
Temporal Evolution

An evolving state space is a functor $\mathcal{X} : \mathcal{T} \rightarrow \mathbf{Adm}$.

Meta-Admissibility

Self-modifications must preserve reachability structure. A transformation $F_{tt'}$ is meta-admissible iff the diagram commutes—meaning it creates no new admissibility distinctions.

The Unified Geometry of Admissibility



Core Synthesis

Observation provides local sections.
Planning requires a global section.
A temporal functor moves through a
lattice, mapping via categories to
resolve the sheaf gluing failure.

The Central Conjecture: Cohomological Obstruction

$$||[\delta s]|| \geq c \cdot \inf_{\phi \text{ consistent}} \mathcal{D}_A(\phi)$$

The Insight

Admissibility distortion is not just a statistical error; it is the first Čech cohomology group $\check{H}^1(\mathcal{U}; \mathcal{F}_A)$ of the **reachability** sheaf.

The Implication

If the first Čech cohomology is **non-trivial**, an inadmissible encoder must exist. The AI planning problem is formally equivalent to bounding and eliminating a topological obstruction.